

New Product Launch Strategy for Network Effects Products

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In recent years, there has been a growing interest in the link between launch strategy decisions and new product performance. Much of that research focuses on investigating successful launch strategies for innovative, high-technology products. With the rapid growth of information technology as one high-technology sector, in certain industries, network effects occur, which change the competitive game. The existing literature offers little decision-making guidance to managers on how to successfully introduce a product that exhibits network effects. The authors discuss the influence of network effects on the dynamics of market competition and on consumers' consumption behaviors. They argue that, because of these changes, the priority of particular performance objectives and the impact of specific launch strategies differ for products that exhibit network effects from what current wisdom and empirical results prescribe. These ideas are formalized in a conceptual framework and a series of research propositions.

Keywords: *network effects; product launch strategy; information technology; high-tech products*

Successful introductions of new products into the market are important for the survival and growth of companies. However, many new products fail, and the failure rate is as high as 50 percent at launch (Cooper and Edgett 1996). The type of launch strategy employed is one of the

key determinants of new product success. A new product could be highly innovative, incorporating advanced technologies that render them superior to competitive products, but still fail due to a poor launch (Campbell 1999).

Although many researchers have focused on identifying the most successful launch strategies (Ali, Krapfel, and LaBahn 1995; Beard and Easingwood 1996; Hultink, Griffin, Hart, and Robben 1997; Traynor and Traynor 1989; Yoon and Lilien 1985), the relationship between launch strategy and new product performance is not yet fully understood. There are characteristics that some product categories exhibit, termed *network effects*, that cause unique competitive dynamics and change customer response patterns. These call for product launch strategies that differ from what might normally be effective. It is possible that the application of commonly held "appropriate" launch strategies to network effects products may, in fact, be detrimental to the marketplace performance of such products.

For example, previous work as well as conventional wisdom suggest that *skimming pricing* and *niche targeting* are successful launch strategies for innovative high-technology products, especially under conditions of low competition, in the early stage of the product life cycle or at the beginning of the diffusion process (Beard and Easingwood 1996; Hultink et al. 1997; Rogers 1995; Yap and Souder 1994). Beard and Easingwood (1996) empirically demonstrated that products are usually launched at a high price when market and technology uncertainty are high. Hultink et al. (1997) showed that skimming pricing, niche market targeting, and new channels are successful launch strategies for certain innovative products. This stream of prior research has sought to empirically assess the impact of innovativeness, market uncertainty, and technology

uncertainty on launch strategies. However, it is not clear that these relationships appear consistently across industries and product categories, even across high-technology industries.

"High-tech" encompasses all advanced technology, including chemical-based, materials-based, biology-based, and electronically based advances, among others. A particular subset, those based on information technology (IT) and communications, typically exhibit *network effects* (Arthur 1996; Katz and Shapiro 1992). This phenomenon is one in which the value of a product to its users increases as more people use the same good (Farrell and Saloner 1985; Katz and Shapiro 1985). Network effects are important in IT markets that rely on communication technology networks and compatible protocols/standards. Common examples of products exhibiting network effects are computer operating systems and compatible software, IT products, telecommunications, and facsimiles (Farrell and Saloner 1986; Gupta, Jain, and Sawhney 1999; Padmanabhan, Rajiv, and Srinivasan 1997; Shurmer 1993).

As the economy becomes more heavily based on IT products and services that exhibit this unique characteristic, it is important to revisit the existing knowledge base grounded in empirical research and to expand that body of knowledge to incorporate this new economic behavior. Casual observation suggests that products exhibiting network effects succeed in a different way. They typically are introduced to the market as broadly as possible and are managed with the objective of rapid market penetration. One of the most common examples is America Online (AOL). AOL's success can largely be attributed to its rapid penetration of the market through a giveaway of its products/services for consumers to try for free for a period of time. This example, and many others, suggests that launch strategy prescriptions cannot be universal across industries, economic environments, and technological environments. These considerations highlight the need for researchers to reexamine launch strategy issues at more fine-grained levels.

The purpose of this article is to (1) investigate how the relationship between launch strategy and new product performance differs for network effects products versus nonnetwork effects products and (2) develop a conceptual framework and research propositions to enrich our understanding of this unique context. We focus on marketing strategies designed to aid product success in a network effects environment. Network effects can be promoted at the technology standard level as well, through the promotion of licensing and joint venture arrangements, cooperative research agreements, and association memberships. In this article, we choose to focus on the consideration of

tactical, product-level issues over which product managers can exercise direct control. The issue of arranging industry agreements to adopt particular design standards, while a critical issue, is beyond our consideration here.

The remainder of this article is organized as follows. We first develop the concept of network effects, specifically with respect to its implications for marketing theory and practice. We address the issue of how network effects influence the dynamics of market competition and consumers' purchasing behavior. Cases and examples are provided to demonstrate how conventional wisdom/traditional approaches (for nonnetwork effects products) may be detrimental to network effects products. We then discuss how these dynamics may affect performance objectives and launch decisions, and we develop a conceptual framework and research propositions with respect to each launch strategy. We hope that this framework can serve to guide future empirical research and, ultimately, managerial decision making in a network effects context. Finally, we close with conclusions and managerial implications.

ELABORATION OF THE NETWORK EFFECTS CONCEPT

Economists have studied network effects for decades (Farrell and Saloner 1985; Katz and Shapiro 1985). In the strategic marketing literature, Dickson, Farris, and Verbeke (2001) identified network effects as one of a number of dynamic feedback mechanisms that have an impact on organizational learning, its resultant evolutionary path, and the development of competitive dynamics in an industry. In this article, we focus on only one of these elements: network effects, considered by Dickson et al. as an asset positioning feedback effects.

The term *network effects* refers to the phenomenon in which the value of a product to one user increases as more users adopt the product (Katz and Shapiro 1985). This characteristic, commonly referred to as "the more, the merrier" (Lee 2000), changes short-term performance objectives for the firm, the dynamics of market competition, and consumption behavior. The guideline for success in a network effects context is maximizing the installed base rapidly rather than skimming marginal profits. Once a technology/product becomes dominant, the process is exceedingly difficult to reverse due to extremely large switching costs that deter consumers from adopting new alternatives even if they are superior (Farrell and Saloner 1986; Redmond 1991). Consumers' expectations about the future installed base and the resultant benefits of "the more, the merrier" phenomenon play a critical role in their product adoption decisions. Network effects emanate from three sources.

Direct Effects—The Relationship of the Product to Its Customer Base

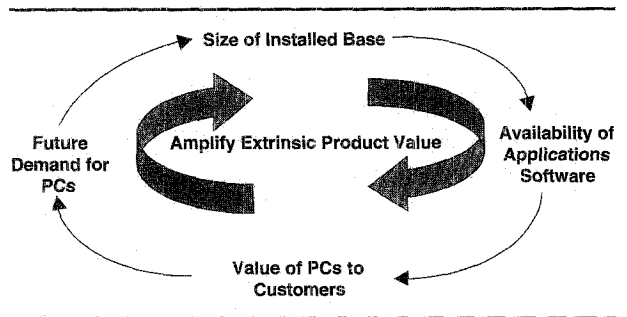
Direct network effects occur when the value of a good to any user is an increasing function of the network's size (Farrell and Saloner 1985; Katz and Shapiro 1985, 1986; Sheremata 1997; Shurmer 1993). The effect is simply generated from the growing number of users adopting the same product. Direct network effects are also called *demand-side economies of scale* (Katz and Shapiro 1986) or *economies of mass adoption* (Norsworthy and Lee 1998). For example, as additional users join a fax machine network, their utility from using the machine increases since each user has a wider pool of potential recipients to send faxes to. Early versions of the videophone and FedEx's Zapmail are examples of products that failed due to their inability to achieve a broad enough installed base early enough for consumers to perceive the benefit.

Indirect Effects—The Relationship of Product Compatibility to Product Utility

Indirect network effects occur when the introduction of complementary goods increases in quantity and decreases in price as consumption of the primary good increases (Arthur 1994; Kahan and Klausner 1997; Riggins, Kriebel, and Mukhopadhyay 1994; Sheremata 1997; Shurmer 1993). Indirect network effects increase with the number of adopters, since the market potential becomes increasingly attractive to firms considering entry with complementary products (Shurmer 1993). Following from this, the more application software (an ancillary good) that is written for an operating system (primary good), the higher is the value for customers who use that operating system.

Direct and indirect network effects affect each other and generate a self-reinforcing character (Hill 1997)—a *positive feedback effect*, as shown in Figure 1. A large installed base of personal computers leads to a greater availability of software applications or other compatible hardware, which in turn draws more users into the network. Hewlett Packard used its printer network effects to market system compatible scanners and PCs. AT&T sold answering machines as add-ons to their phones. This availability of compatible goods has a positive impact on the value of a particular machine to consumers and leads to greater demand for them. This translates into a greater installed base. Network effects products become more valuable to current or potential customers as their use becomes more widespread (direct network effects) (Kahan and Klausner 1997). The enhanced value is *extrinsic* to the physical features of the product itself; it is derived from the increased availability and reduced costs of complementary/compatible products.

FIGURE 1
Positive Feedback of Network Effects in the PC Industry



SOURCE: Adapted from Hill (1997).

The Standards Issue

Standardization or the use of design protocols is also one of the sources of network effects. Standardization feeds the reinforcing cycle between primary and ancillary products, since compatibility is normally maintained by adhering to a common technological standard. A standard with a large installed base will attract more complementary products and help convince consumers that they will not become locked into a poorly supported design (Shurmer 1993). For example, as the number of users of VHS video recorders increased, so did the supply of prerecorded cassettes. Matsushita dominated the VCR market because its VHS videocassette recorder emerged as the dominant design over Sony's alternative Betamax format.

THE IMPACT OF NETWORK EFFECTS ON MARKET COMPETITION AND CONSUMPTION BEHAVIOR

We now turn to a consideration of the impact that a network effects product has, due to its unique characteristics, on how consumers perceive value in products and on how the dynamics of competition unfold. Each of these dynamics aids in understanding the appropriate launch objectives for a network effects product and thereby helps determine the key launch strategies to be used.

Lock-In and Winner-Take-All Competition

The positive feedback effect described earlier generates a self-reinforcing character of network effects that makes strong competitors get stronger and any follow-on competitors grow weaker (Shapiro and Varian 1999b). In its most extreme form, positive feedback can lead to a

winner-take-all market, in which a single firm or technology vanquishes all others. Thus, firms compete, in the context of network effects, by attempting to expand the scope of ancillary products to leverage the installed base of primary products, thereby activating positive feedback effects (Shapiro and Varian 1999b). In the mid-1980s, Novell introduced its network operating system, NetWare, as a way of connecting personal computers in local networks. Novell managed these cross-product (between primary product and ancillary products) development relationships very proactively by heavily discounting NetWare to penetrate the market and by offering incentives for software developers to write for NetWare (Hill 1997).

Microsoft has managed the development of its operating system as an "open" system, in the sense that independent software vendors have full access to the Application Program Interfaces in order to make their applications work well with Windows (Shapiro and Varian 1999a). By building on Window's success, the firm ensured its own. In contrast, although Apple was first to the PC market, the installed base of Apple machines grew more slowly than the installed base of Wintel machines (Hill 1997; Sheremata 1997). Yet another example of a winner-take-all competitive dynamic is Matsushita's VHS format, after a period of competition with Sony's Betamax format, becoming the industry standard in the VCR market (Chen 1997; Cusumano, Mylonadis, and Rosenbloom 1992; Rosenbloom and Cusumano 1987; Shapiro and Varian 1999b; Sheremata 1997).

Clearly, new standards eventually overtake old ones, as digital video discs (DVDs) are currently replacing conventional videos in the movie rental market. Hence, "lock-in" is not a permanent effect. But it takes considerable intrinsic product advantage and additional market development investment to break the lock-in cycle, such as a radical innovation that provides extremely superior performance (Leifer, O'Connor, and Rice 2000). In addition, with network effect products, the new product/technology must create enough extrinsic value to trigger positive feedback effects and establish a large installed base to compete with the incumbent's network. Thus, DVD players cannot overcome the VHS market without enormous industry investment in producing movie titles on DVDs and other complementary products.

Consumption Behaviors Specific to Network Effects Products

It is useful to consider how the sources of customer value differ for network effects and nonnetwork effects products. *Intrinsic* product value refers to the features/attributes designed into the product itself as well as experiences derived from the augmented product. For example, fax machines have a certain number of attributes/

functions, including a particular resolution, speed of information transfer, and ability to save data, as well as after-sales support and maintenance packages.

However, if only one person has a fax machine, it is useless despite how well designed the features or how extensive the postsales support. The more people that use fax machines with the same design protocol, the more useful the fax machine is to each of them. This is the second source of value, labeled *extrinsic* value, and it is unique to network effects products. Extrinsic value is the set of benefits derived from outside the product itself, such as the size of the installed base and the availability of compatible and complementary products that enable greater use of the base product. The intrinsic value of a product is constant, but extrinsic value varies in the network effects context relative to the size of the installed base.

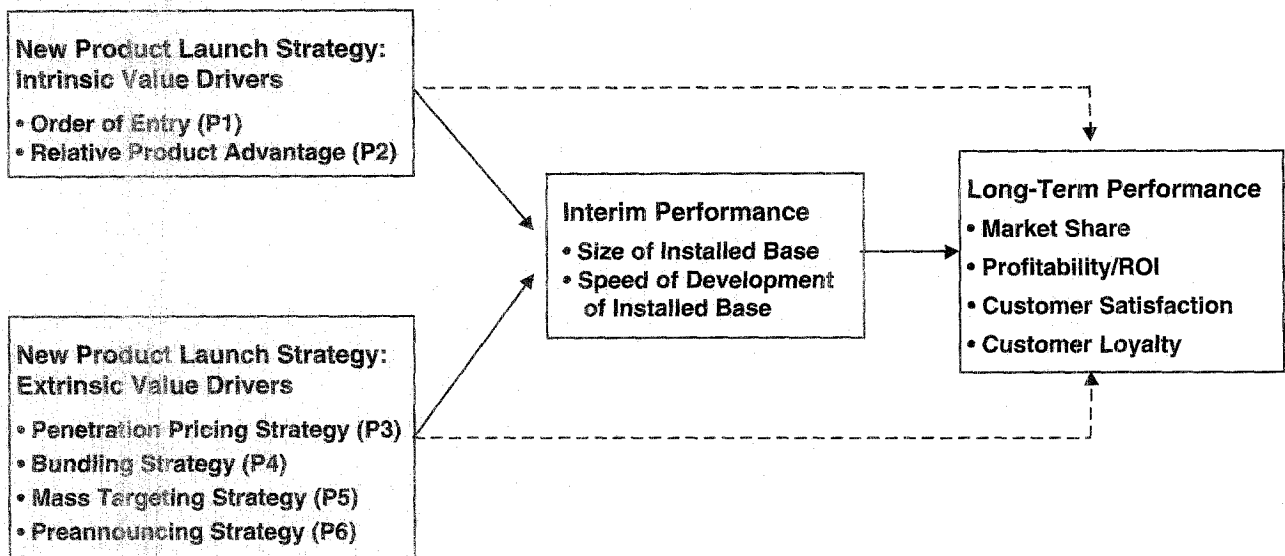
In contrast to consumers' experiences with nonnetwork effects products, customers benefit from a network effects product not only as a result of the *intrinsic* product value (its features) but also from its *extrinsic* value (its links to other products and users). Thus, the size of the installed base is a critical determinant of success for a network effects product. Because of positive feedback effects, whichever brand gets ahead attracts more support from ancillary product suppliers, and this in turn helps fuel further growth. Consumers' purchasing decisions are guided not only by a product's current installed base but also by their expectations of the size of the future installed base (Kahan and Klausner 1997; Katz and Shapiro 1985, 1986; Sheremata 1997; Shurmer 1993). Thus, marketing strategies designed to influence customer expectations are critical in the context of network effects (Sheremata 1997).

CONCEPTUAL FRAMEWORK AND RESEARCH PROPOSITIONS

Figure 2 presents a conceptual framework delineating the relationships between product launch strategy and performance. This framework extends previous work on launch strategy by incorporating considerations unique to product categories characterized by network effects. It illustrates how various facets of a launch strategy affect intrinsic and extrinsic product value and thus influence new product performance in the context of network effects. Considering the role of network effects played in market competition and consumers' consumption behaviors, we suggest that increasing intrinsic product value attributable to order of entry (which can lead to economies of scale and therefore low cost) and relative product advantage is important but not sufficient to achieve superior long-term performance in the context of network effects. It is proposed that the success of network effects products relies heavily on the maximization of extrinsic product value in addition to the influence of intrinsic

FIGURE 2

Product Launch Strategy and Performance of Network Effects Products: A Conceptual Model



NOTE: ROI = return on investment.

product value. Marketing strategies that can be applied to the maximization of extrinsic product value include penetration pricing, bundling, mass targeting, and preannouncement strategy as opposed to skimming pricing, unbundling, niche targeting, and no pre-announcement, as depicted in Figure 2.

The model proposes that the impact of various marketing launch strategies on new product performance will differ in a network effects context from current prescriptions in the literature regarding launch strategy for high-tech products. In discussing this framework, we first introduce the appropriate measures of the dependent variable—new product performance. Second, we discuss specific launch strategies and offer a series of research prescriptions regarding each strategy's relationship to superior performance in a network effects context. We first propose that two launch strategies associated with the improvement of *intrinsic* product value are not sufficient to succeed in the context of network effects. Then, we elaborate on four launch strategies conducive to increasing *extrinsic* product value that, we posit, are key to success in the context of network effects.

New Product Performance

Researchers suggest the use of multidimensional measures to capture the overall performance of a new product at the aggregate level (Griffin and Page 1993, 1996; Kerin, Varadarajan, and Peterson 1992). Measures typically address the degree to which organizational goals involving new product-market, customer, and financial objectives

are achieved. Differentiating between short-term and long-term measures is also important, particularly in a network effects environment. Initially, firms that introduce network effects products compete to increase the size of their installed base (Arthur 1994; David and Greenstein 1990; Farrell and Saloner 1986; Hill 1997; Katz and Shapiro 1985; Shapiro and Varian 1999b). Increasing the installed base will enhance customer satisfaction, because they experience extrinsic value and because they expect the installed base to prevail in the competitive marketplace (Shapiro and Varian 1999b). It is because of the establishment of a large installed base that a network effects product can lock in customers even with an inferior technology (Katz and Shapiro 1986). Thus, a large installed base becomes the critical initial performance objective in the short run.

This is a sequential path for long-term success—from maximizing the installed base to long-term new product performance on multiple dimensions, as shown in Figure 2. With a large installed base, the dominant design has the market power to charge a higher price, thus enjoying a higher profit margin. Being locked in, customers are willing to pay a price premium for a dominant design (Brynjolfsson and Kemerer 1996; Gandal 1994; Hartman and Teece 1990). The overall performance measures, then, include three components: (a) market share objectives—size of the installed base,¹ speed with which the installed base is developed, and long-term market share; (b) customer objectives—customer satisfaction and customer loyalty; and (c) financial objectives—long-term profitability and return on investment.

Order of Entry

Pioneering advantages. Several studies have shown that pioneers have long-lived market share advantages, and they often become market leaders (Dyer, Gupta, and Willemo 1999; Golder and Tellis 1993; Green, Barclay, and Ryans 1995; Lieberman and Montgomery 1998; Robinson and Fornell 1985; Urban, Carter, Gaskin, and Mucha 1986). Pioneering advantage derives from several sources. First are *product-based advantages*, which result from the large volume supply of the product (Bain 1956). Economies of scale and learning curve effects are product-based advantages that lead to superior cost positions for pioneers (Lilly and Krishnan 1996; Robinson and Fornell 1985; Urban et al. 1986). Product-based advantages build barriers to entry against later entrants (Bain 1956). Other important product-related advantages enjoyed by first entrants include technology leadership (Gilbert and Newbery 1982; Lieberman and Montgomery 1988) and preemption of scarce assets (Lieberman and Montgomery 1988; Prescott and Visscher 1977; Spence 1977). In general, pioneers tend to provide superior products; broader product lines; and ultimately, lower costs than late entrants, and therefore pioneers often outperform followers (Kerin et al. 1992; Lambkin 1988; Robinson and Fornell 1985).

A second source of advantage for market pioneers is a set of *consumer-based advantages*. These are benefits derived from the way consumers first choose and then repurchase products (Golder and Tellis 1993). Consumers tend to develop stable preferences for early entrants, especially when product-related information is imperfectly available (Bain 1956; Lieberman and Montgomery 1988; Schmalensee 1982), as is the case with novel, high-technology products. Under uncertainty, buyers remain loyal to the pioneer's brand as it is the only one they know. Pioneers may also influence consumers' evaluation of the attributes in the product category and may therefore have an increased likelihood of becoming the standard for the product category (Carpenter and Nakamoto 1989). Finally, pioneer advantages may also arise from buyer switching costs. With switching costs, late entrants must invest extra resources to attract customers away from the pioneer (Lieberman and Montgomery 1988).

The importance of order of entry effects in the context of network effects. As discussed above, being first to the market is, for the most part, associated with superior performance in the literature. However, in the context of network effects, these effects may not be so strong, since network effects alter the dynamics of market competition and consumers' consumption behavior (Farrell and Saloner 1985, 1986; Katz and Shapiro 1985, 1986; Redmond 1991; Shapiro and Varian 1999b). Since consumers' purchasing decisions depend on both *intrinsic* and *extrinsic* product

value derived from the size of the network, even if a product is late to the market, as long as a large installed base can be established quickly, network effects may override first-mover advantages. In fact, Tellis and Golder (1996, 2001) showed that market pioneers do not always enjoy an advantage if a follower commits more resources to establishing large distribution and therefore rapidly builds an installed base (Tellis and Golder 1996, 2001).

Pioneers have great potential to gain market acceptance (pioneers tend to achieve substantially higher market share than do late entrants) because of the product-based, customer-based, cost, and differentiation advantages mentioned previously. Among those advantages, switching costs (Lieberman and Montgomery 1988) and customer preference formation (Carpenter and Nakamoto 1989) are the most critical factors that positively affect the development of market acceptance. However, in the presence of network effects, both switching costs and customer preference formation may depend more on the size of the installed base and customer expectations regarding its size in the future than on the order of entry itself.

Pioneers with an insufficient installed base may potentially lose the market to quick followers who rapidly penetrate the market and establish an overwhelming installed base. Lambkin's (1992) empirical results demonstrate the risks facing market pioneers that fail to enter their markets on a large enough scale. Pioneering firms that have built their advantage around technology-specific knowledge and manufacturing assets are at a disadvantage to comparable followers who exhibit distribution and brand equity advantages, because the latter are easier to redeploy (Dickson et al. 2001). In addition, laggards can "free ride" on a pioneer's investments in R&D, buyer education, and infrastructure development (Lieberman and Montgomery 1988). Thus, laggards with a large market presence (installed base), which implies substantial assets and capability, can imitate a pioneer/innovator and dominate the market (Nelson and Winter 1982). For example, Apple was a pioneer and introduced the Mac PC in 1977. However, IBM quickly dominated the business market, using marketing clout and product innovation (Schnaars 1986). Prodigy was first into the online services market but was passive in building its subscriber base to take advantage of network effects. As a result, it quickly fell from its leading position (Arthur 1996).

In summary, the long-term market share of a network effects product may not depend on order of entry so much as on other marketing strategy factors that affect the size of the installed base in the short term. Being first to the market is not sufficient, in itself, to win in a network effects context (Arthur 1996; Cusamano et al. 1992; Hill 1997; Katz and Shapiro 1992; Shapiro and Varian 1999b). Pioneers in network effects contexts may still lose the game if the installed base is not big enough immediately. There-

fore, we propose that pioneers, on average, tend to outperform later entrants. However, when network effects exist, the impact of launch strategy decisions that can rapidly maximize the installed base may override the impact of pioneer advantage on new product performance. In other words, order-of-entry effects are less influential in a network effects world than in a nonnetwork effects world. Thus, we propose the following:

Proposition 1: For network effects products, the positive impact of order of entry on new product performance (the extent of intrinsic product value) can be overridden by the positive feedback effects derived from the development of a large installed base (the extent of extrinsic product value).²

Product Advantage Strategy

Product advantage strategy and performance. Product advantage refers to a bundle of tangible and intangible benefits that a product offers to its customers that is unique and superior to competitive brands (Cooper 1979; Cooper and Kleinschmidt 1987, 1990; Li and Calantone 1998). Product advantage may be based on product image (reputation) (Robertson and Gatignon 1986; Traynor and Traynor 1989), product quality/reliability (Li and Calantone 1998), perceived product value (Green, Gavin, and Aiman-Smith 1995), or technological innovativeness (Beard and Easingwood 1996; Cooper and Brentani 1991; Hultink et al. 1997). It is central to the definition of a high-technology product.

Product advantage is not only positively associated with new product performance (Li and Calantone 1998) but also increases the adoption rate in the commercialization stage (Rogers 1995). Empirical evidence suggests that relative product advantage is one of the most important determinants of new product trial and adoption. Holak and Lehmann (1990) found that, for an array of durable goods, product advantage is positively associated with consumer purchase intention. Li and Calantone (1998) surveyed 236 software companies and found that new product advantage positively affects product market performance. Green, Gavin, et al. (1995) also found that the higher the perceived value of the product, the greater the long-term performance of the product.

The role of product advantage strategy in the context of network effects. It is not clear, however, whether product advantage plays a similar role for network effects products. The sales potential of a new product depends on the benefits the product provides to consumers (Horsky 1990). For nonnetwork effects products, product advantage derives from the intrinsic features of the product—the embodied product functions, while for network effects products, advantage derives from both intrinsic and extrin-

sic benefits—the extra value that is associated with the increase in the size of the network (Shurmer 1993). The economics literature suggests that a network effects product emerges as a de facto standard or dominant design due to the early success it has in building a large user base and in promoting expectations about its future installed base (David 1985, 1987; David and Greenstein 1990; Farrell and Saloner 1985, 1986; Katz and Shapiro 1985, 1986, 1992; Shapiro and Varian 1999b; Shurmer 1993). Competition is based on a standards war or the size of the network that ultimately locks in customers, rather than on product advantage alone.

Several real-life examples indicate that inferior products and technologies may, in fact, become the industry standard. The classic example of a market's lock-in to an inferior technology is the QWERTY format for typewriter (and now computer) keyboards. This initial keyboard layout has penetrated the market so heavily that the superior Dvorak design could not make inroads, although it is a much more efficient keystroke system (Hill 1997; Redmond 1991; Shapiro and Varian 1999a). Similarly, some contend that Microsoft's Windows operating system dominates the PC market not because of its superior technology/product advantage but because its rapid penetration strategy locked in the market and became the industry standard (Hill 1997; Shapiro and Varian 1999a). Viewed in this way, Apple's strategy of penetrating the student market with the expectations that young people would grow up and insist on Apple machines in their offices and executive suites was flawed. IBM had already locked in the business office market, and many new employees have been forced to adopt the IBM system, even though they may have a preference for Apple-based systems. Finally, the 1950s programming language FORTRAN dominated ALGOL not because FORTRAN is superior to ALGOL but because FORTRAN was more widely promoted and taught in universities and industry in the 1960s than other programs (Arthur 1994; Dickson 1995). These examples indicate that when network effects exist, there is little chance to win on the basis of product advantage alone if the market has been locked into a dominant design. What emerges as the dominant design is not necessarily the best available technological solution (Arthur 1994, 1996; Hill 1997; Redmond 1991; Shapiro and Varian 1999a, 1999b). In the context of network effects, factors other than product advantage also significantly influence a product's success.

We expect that product advantage, on average, will increase new product performance. However, in the context of network effects, the impact of launch strategy decisions that can rapidly maximize the installed base may override the impact of product advantage alone on new product performance (Hartman and Teece 1990), especially at the early stage of the product life cycle when the development

of the installed base has just begun. Thus, we propose the following:

Proposition 2: For network effects products, the positive impact of product advantage strategy on new product performance (the extent of intrinsic product value) can be overridden by the positive feedback effects derived from a large installed base (the extent of extrinsic product value).

Relative Impact of Launch Strategies on Extrinsic Value

The discussion pertaining to Propositions 1 and 2 suggests that order of entry and product advantage strategy are critical for nonnetwork effects products' success but may not be sufficient for survival in the world of network effects products. The marketplace is filled with examples of firms that provide less than top quality in initial product releases to gain a foothold in the market and then make up for the quality differential through provision of upgrades in later periods (Padmanabhan et al. 1997).

Arthur (1994) proposed that the reason that inferior technology often becomes the accepted standard depends on historically small (i.e., seemingly insignificant) events that are path dependent and so drive the course of technology advancement.³ We believe this winner-take-all phenomenon is more a strategic issue rather than just an occurrence that happens via small historical events. Educating consumers regarding the product's features and functionality, and how to fully benefit from those; building an open-systems strategy to encourage creation of ancillary products; and creating expectations in consumers' minds that this product will become the standard are all managerially controllable practices. These activities are not dependent on insignificant events but rather on clear strategic direction, understanding of market processes, and on appropriate allocation of resources to aid in creating market presence. We believe that it is the aggressive use of launch strategies that determines the success of a network effects product rather than a reliance on small historical events. Marketing strategies designed to influence consumer expectations and leverage the positive feedback of network effects are critical to success in the context of network effects.

We now proceed to launch strategies that we expect will be most effective in developing the market for network effects products. Here, we focus on pricing, bundling, targeting, and preannouncement elements of launch strategy. For each element, we consider whether the prescriptions offered in the literature for high-tech markets make sense given the uniqueness of that subset of high-technology industries that exhibit network effects.

Pricing Strategy

Price is one of the major determinants of buyer choice and also one of the most important elements determining company market share and profitability (Kotler 1994). Companies handle pricing in a variety of ways. The appropriate pricing strategies for high-tech products as generally acknowledged in the literature may be detrimental for those exhibiting network effects.

Pricing strategy for high-tech products without network effects. When introducing an innovative or technology-based product, sellers have traditionally relied on a *skimming pricing strategy*—initially setting the price high to “skim” profits from less price-sensitive innovators and early adopters and then decreasing prices gradually to reach broader markets (Dorward 1987; Grunenwald and Vernon 1988; Higgins and Shanklin 1992; Kotler 1994; Robertson 1993; Tellis 1986; Winkler 1984). Results from empirical studies indicate that skimming pricing has a positive impact on new product performance for innovative products (Beard and Easingwood 1996; Hultink and Schoormans 1995; Yoon and Lilien 1985). From the *producers' perspective*, skimming pricing is typically applied to new products with certain unique characteristics (e.g., patent protection) that guard them from competition in the initial stages of market development (Dorward 1987). The firm charges a high price in order to recover as much profit as possible before competitors enter the market (Kotler 1994). Price sensitivity is lowest during early phases of the life cycle (Dorward 1987; Nagle 1987) when early adopters and innovators are purchasing the product (Moore, Boulding, and Goostein 1991; Rogers 1995). From the *consumers' perspective*, price provides a source of information for buyers to judge product quality (Carpenter, Glazer, and Nakamoto 1994; Sivakumar and Raj 1997), especially in cases where they are purchasing highly innovative products and lack the knowledge required to objectively evaluate the product/technology.

Pricing strategy for network effects products. The subset of high-tech products that exhibit network effects (e.g., IT) requires large investments in complementary infrastructures to be commercially viable or to ensure that multiple standards are not competing in the market (Bulte 2000; Cusumano et al. 1992). Because of the presence of “the more, the merrier” dynamic, the objective of a network effects product is not to make profits in the early stage of product life cycle but to maximize the installed base of the launched product quickly. Thus, penetration pricing has a critical place in the context of network effects. A penetration strategy is favored when a firm attempts to exploit scale or experience economies (Tellis 1986), to reduce market penetration cycle time (Robertson 1993), and to grow quickly (Arthur 1996; Hill 1997;

Grunenwald and Vernon 1988; Robinson and Lakhani 1975). In the presence of network effects, a firm may strategically offer a price performance advantage at the early stage of the product life cycle by penetration pricing to establish a large network (Hartman and Teece 1990), create expectations of future large markets, and deter competitors' entry (Katz and Shapiro 1985, 1986). In the long term, however, once the installed base has been established and customers are "locked in," the de facto standard or the dominant design has the market power to charge higher prices and enjoy higher margins. Customers, who derive extrinsic value from the product later in the life cycle and are unwilling to bear the switching costs to alternative networks, could be willing to pay significant price premiums for a dominant design (Brynjolfsson and Kemerer 1996; Gandal 1994; Hartman and Teece 1990).

Price incentives can be offered to initial adopters to attract a stream of future adopters, each of whom contributes to the increase in direct effects benefits for all users and induces additional adopters (Dhebar and Oren 1986). Netscape, for example, pursued a penetration strategy in its attempt to establish its Web browser as the de facto standard for enabling commerce on the World Wide Web by giving away the browser for free (Arthur 1996; Hill 1997). More recently, Linux is threatening Microsoft's dominant position through its use of open source strategy (Kerstetter, Hamm, and Ante 2003). In their study of the fax machine market, Economides and Himmelberg (1995) found that penetration pricing positively affects the expansion of the network. Therefore, in the presence of network effects, we expect to see that launching a new product with penetration pricing will lead to high performance.

Proposition 3: For network effects products, a penetration pricing strategy during product launch is more conducive to superior new product performance than a skimming pricing strategy due to its influence on extrinsic product value.

Bundling Strategy

A bundling strategy involves selling two or more products as a set for a single price (Kotler 1994; Mulhern and Leone 1991; Schmalensee 1984; Yadav 1994; Yadav and Monroe 1993). A firm may use bundling for various purposes: to reduce costs, to expand the market, and to improve product performance (Eppen, Hanson, and Martin 1991). In addition, especially for high-tech products, bundling can be used to reduce consumers' risk perception of a new product. Consumers who have little knowledge about the new product may view purchasing a bundled offering as less risky because components of the bundle will work together properly and be covered by a single warranty (Paun 1993). For example, Microsoft offered its

Office software suite that bundled Excel spreadsheet, Word processing, Access database, and Power Point presentation software together. Consumers (except, perhaps, experts who are knowledgeable users and may care more about the specific features of the product rather than ease of use) feel more comfortable buying the Office package than choosing spreadsheet, word processing, database, and presentation software from different producers (e.g., Lotus, Word Perfect, and DB2). In some cases, firms even bundle new products as free gifts, which offers customers a riskless trial opportunity.

The role of bundling strategy for network effects products. In the context of network effects, the rationale behind product bundling is that it increases firm performance through increasing competitive advantage via the *multiplier effect*, defined as the incremental volume of a primary product's sales that the bundle's complementary products generate (Sengupta 1998).

With the presence of network effects, compatibility is important, and the value to a consumer of owning a product is an increasing function of the installed base of that product (direct network effects that increase extrinsic value) as well as the availability of other compatible products (indirect network effects that increase extrinsic value) (Arthur 1994; Katz and Shapiro 1985, 1986, 1992; Sheremata 1997; Shurmer 1993). A bundling strategy can be used by the firm to link the primary product with other compatible ancillary products, reinforcing positive feedback and thereby increasing the demand for both. Bundling is also used to leverage the product from an older version to an upgrade version. The logic is to offer customers a smooth migration path with backward compatibility (Shapiro and Varian 1999b). For example, Microsoft leveraged its customer base of 60-million DOS users onto Windows, to Windows95, and then to Microsoft Network by offering inexpensive upgrades and by bundling applications such as Internet Explorer (Arthur 1996). In summary, the advantage of bundling strategy for network effects products is to expand the installed base through leveraging positive feedback effects, which increases the extrinsic value of the product to any single user in the network. Therefore, we propose the following:

Proposition 4: For network effects products, a bundled product launch strategy will be more conducive to superior new product performance than an unbundled product launch strategy because it leverages positive feedback effects, thereby adding to the extrinsic value of the product.

Targeting Strategy

Targeting strategy for general high-tech products. Research suggests that *niche targeting* is most appropriate for

innovative high-tech products (Beard and Easingwood 1996; Hultink et al. 1997; Moore 1991). A niche market is a narrowly defined group that may seek a special combination of benefits for which it is willing to pay a premium. In theory, the diffusion of an innovation proceeds from the small segment of the market referred to as innovators, to early adopters, and finally to early and late majority markets (Moore 1991; Rogers 1995). Thus, targeting innovators is an appropriate tactic for technically complex, highly innovative products, especially at the early stage of the product life cycle (Beard and Easingwood 1996). This segment is highly responsive to the benefits of a novel product, and in general, they are opinion leaders who influence later adopters (Beard and Easingwood 1996; Rogers 1995). In conventional markets, niche market targeting is then favored by followers as the market continues to segment into finer and finer subsets and smaller or follower firms seek to avoid direct competition with market leaders (Kotler 1994; Lambkin 1988).

Targeting strategy for network effects products. For products characterized by network effects, the major reason to target broadly is to quickly expand the installed base and lock in the mass market (Shapiro and Varian 1999b). America Online built up a lead of more than 4.5 million subscribers initially by giving away free services to the entire potential market (Arthur 1996).

Mass targeting is commonly used (by pioneers or large firms) to achieve economies of scale, both in production and distribution, resulting in a cost advantage and in the preemption of future competition (Biggadike 1979; Holak and Lehmann 1990; Kalyanarm and Urban 1992; Lambkin 1988, 1992). In the context of network effects, however, mass-market strategy is not motivated by production scale economies (since marginal cost of production is extremely low at any volume) but by the need to create extrinsic value for customers. Hill (1997) suggested the need to build wide initial distribution to start the positive feedback of network effects. The rapid growth of the installed base leads to increased value perceived by any single user in the network.

Moore (1991) proposed that a chasm exists between the innovator/early adopters and the early majority in the world of high-tech products because early adopters have little influence on the adoption patterns of the majority. Therefore, managerial attention to the expansion of the network is key, as conventional diffusion patterns may not hold. When network effects are present, then, mass market targeting strategy is expected to be more successful than niche targeting strategy due to its positive impact on extrinsic product value.

Proposition 5: For network effects products, a mass targeting strategy during product launch is more conducive to superior new product performance than a

niche targeting strategy due to its positive impact on extrinsic product value.

Preannouncement Strategy

Preannouncement is a formal and deliberate communication before an actual new product introduction (Eliashberg and Robertson 1988; Robertson, Eliashberg, and Rymon 1995). A preannouncement may be directed at competitors or at customers. In preannouncing to *competitors*, the logic is that by publicly announcing intentions early, competitors will be discouraged from entering the market. The preannouncing firm benefits from its accessibility to an efficient distribution system (Robinson and Fornell 1985) and the creation of barriers to entry for other firms by leaving them the unprofitable segments (Schmalensee 1982).

Another important reason for preannouncing new products, however, is to influence potential *customers* (Eliashberg and Robertson 1988; Rabino and Moore 1989). Preannouncing before market introduction can build buyer curiosity and interest in the new product (Lilly and Krishnan 1996; Ziegler 1995) and can also encourage buyers to delay their purchases until the announcing firm's product is available (Lilly and Walters 1997).

Reasons for preannouncing for network effects products. When introducing a network effects product, preannouncing can lead to higher performance than not preannouncing for the following reasons. First, preemption is one of the most crucial marketing strategies used in standards battles (Shapiro and Varian 1999a). The logic of preemption is to build an early lead to stimulate the positive feedback of network effects (Shapiro and Varian 1999b). Strong network effects reinforce the effectiveness of preemptive announcements. Rivals will retreat in a market not only if the market is "locked in" but also if they believe it will continue to be locked in by a competitor (Arthur 1996).

Second, a preannouncement can also be sent for cooperative purposes—seeking alliances or encouraging complementary product design (Lilly and Walters 1997; Robertson 1993; Robertson et al. 1995). These actions stimulate indirect network effects. The prevalence of most successful network products depends on the availability of complementary products (Arthur 1996; Farrell and Saloner 1985; Katz and Shapiro 1985). Both preemption and complementary product stimulation may encourage competitors to follow a particular product standard in the context of network effects.

The third reason is expectations management. Consumer expectations are a major factor in their decisions to purchase a new technology (Holak and Lehmann 1987; Shapiro and Varian 1999b; Sheremata 1997). In the context of network effects, consumers' adoption decisions are

guided not only by a product's current installed base but also by expectations of the future installed base (Kahan and Klausner 1997; Katz and Shapiro 1985, 1986; Sheremata 1997; Shurmer 1993). Because the extrinsic value of a network effects product to its users depends strongly on the size of the installed base, a consumer in the market today also cares about the future success of the product. If consumers expect the product to become popular, the network will grow relatively large. When network effects exist, the strategic reason for preannouncing is to gain a faster takeoff by managing consumers' expectations. Preannouncement can be used as a psychological positioning strategy to convince consumers that the product will become the dominant standard (Arthur 1996). Therefore, potential customers will postpone their purchase of rivals' products and look forward to the introduction of the preannounced product (Farrell and Saloner 1986; Lilly and Walters 1997). Software companies often preannounce "vaporware" to freeze rivals' sales and to influence consumer's expectations (Bayus, Jain, and Rao 2001; Robertson et al. 1995; Shapiro and Varian 1999b).

In summary, a network effects product will benefit from preannouncement more than will a nonnetwork effects product because preannouncement can improve the extrinsic value of the product through (1) increased competitive advantage by preempting the market, (2) stimulating the supply of complementary products, (3) encouraging competitors to follow a particular standard, and (4) influencing buyer behavior by managing their expectations. All of this is predicated on the firm actually delivering on its promises. Clearly, past history affects consumers' and trade members' expectations as well. Therefore, we propose the following:

Proposition 6: For network effects products, preannouncing prior to product launch is more conducive to superior new product performance than not preannouncing, due to its positive influence on the product's extrinsic value perceived by customers.

IMPLICATIONS FOR PRACTICE AND RESEARCH

In this article, we attempt to fill an important gap in the product launch strategy literature. In recent years, many marketing scholars have expressed a considerable degree of interest in the commercialization of high-tech products, defined as those based on advanced technology platforms. A subset of these, based on information and communications technologies, exhibit network effects. A topic of concern to both academics and practitioners is the issue of whether the launch strategies prescribed based on previous work and conventional wisdom are appropriate for network effects products. From a theoretical perspective, we

find that network effects products succeed in a completely different way.

Building on the current launch strategy literature, the conceptual framework presented in this article extends our view of launch strategy decisions in a new direction, to incorporate the economics of network effects. We draw from economic theory to examine conceptually how network effects change market competition, influence consumption behavior, and affect launch strategy for superior new product performance. In contrast to nonnetwork effects products, the success of network effects products depends not only on the product's *intrinsic value*—based on its features, but also on its *extrinsic value*—derived from the size of the network and its connection to other users and complementary products. This implies, then, that conventional wisdom that focuses on increasing the intrinsic product value alone is insufficient to win in the context of network effects. Instead, the growth of the installed base (which increases the extrinsic product value) is the first-priority performance objective in the short term. It is a large installed base that attracts an extensive supply of complementary/compatible products, thus stimulating positive feedback effects between direct and indirect network effects and enabling a network effects product to lock in a customer base even, on occasion, with inferior technology. The characteristics of network effects cause the market to "tip" toward one system—"the tendency of one system to pull away from its rivals in popularity once it has gained an initial edge" (Katz and Shapiro 1994:106). Once a network effects product attracts a sufficient installed base and becomes the dominant design, it will lock in the market and enjoy long-term success (e.g., higher profit margins and greater customer satisfaction).

Furthermore, in a network effects world, being first to the market is advantageous, but the use of appropriate launch strategies to establish the installed base (which increases the extrinsic value of the product) may override the impact of entry timing on performance. Being first to the market is not sufficient, in itself, to win in a network effects context. Pioneers with an insufficient installed base may lose market share to early followers who rapidly penetrate the market and establish an overwhelming installed base.

Similarly, in a network effects environment, while relative product advantage is important, in the short term, consumers' purchase decisions may depend more on the extrinsic value they gain from the size of the installed base rather than on the product's intrinsic features. Once they are locked in to a design that ends up not being adopted as the standard, their switching cost may be substantial. Thus, it could be detrimental for product managers to heavily invest in improving product features at the expense of efforts necessary to establish the installed base.

Given this focus on increasing extrinsic value, then, marketing tactics designed to motivate consumers to

purchase and use early and often appear, theoretically, to be key to superior performance in the long run. While many of these tactics are used in new product launches to stimulate early demand, they are not typically associated with long-term success in and of themselves. But in a network effects environment, they are critical entry levers to even begin playing the competitive game. Customer and competitive dynamics are different in a network effects world, and marketers must be aware of the implications for managerial decision-making purposes.

For managers who are responsible for new product launches, the framework developed in this article offers assistance in analyzing the dynamics of market competition in the presence of network effects, identifying appropriate performance measures, and considering appropriate launch strategies for the new product. Managers should take care to consider the differences in launch strategies that may be necessary when introducing network effects products, since the dynamics of market competition and consumption behavior are so different from those for products without network effects. The proposed conceptual framework suggests that the positive feedback of network effects can be activated and then amplified through a set of aggressive launch strategies. This rationale is parallel to Besen and Johnson's (1986) case study work in the telecommunication sector that shows that firms who use aggressive marketing communication tools can, in fact, overcome current industry standardization agreements and simulate a new round of positive feedback effect in concert with their new standard.

The framework provides some critical elements for consideration when launching network effects products. We have reviewed conventional strategies for marketing of high-tech products. While some of them appear to be appropriate for the network effects environment (i.e., bundling strategy and preannouncing strategy), others may be detrimental. Skimming pricing, niche targeting, and secrecy prior to introduction are all strategies commonly viewed as important for high-tech products. We have built the case in this article, however, that these strategies are not necessarily appropriate in all environments. In fact, they may be detrimental in a network effects environment. In some cases, the tried and true launch strategy prescriptions for new products run counter to what economic theory regarding a network effects environment would suggest.

For researchers, the groundwork is laid in this conceptual model to begin empirical testing of the impact of network effects product characteristics on the relationships between marketing strategies and new product performance. Figure 2 and Propositions 1-6 suggest that the indirect paths from strategy to long-term performance (indicated by the solid lines) are more important than the direct paths (indicated by the dotted lines) in determining a product's long-term success in the context of network effects.

An important contribution of this work is the thorough description of network effects from a marketing perspective. Departing from Arthur's (1994) Path Dependency theory, we suggest that the lock-in phenomenon is more a strategic issue than just an occurrence that happens via small historical events. Firms should take "proactive" actions to establish a sufficient installed base to influence customer preferences rather than let the accidental historical events decide the dominance of a network effects product. Our consideration of the nature of market competition and buyer behavior leads us to believe that the factors under the control of a marketing or product manager are even more critical to a network effects product's success than they may be for nonnetwork effects products.

NOTES

1. The size of the installed base is usually measured by accumulated sales (Brynjolfsson and Kemerer 1996) or accumulated market share (Clark 1999). The speed to establish the installed base can be estimated by *time to take off*—the time required for a product to transition from the introduction stage to the growth stage (Golder and Tellis 1997), or *market penetration cycle time*—the time it takes for a product to reach maximum sales potential (Robertson 1993).
2. We do not mean to communicate that being a first mover is not important. Being a first mover in conjunction with a focus on building a large installed base is potentially the most effective strategy in a network effects environment. The point of the proposition is simply to assert that the relative focus on the installed base may be more influential on new product performance than being a first mover in this context.
3. Arthur (1994) defined *historical small events* as those events or conditions that are outside the knowledge of the observer (e.g., consumer preference of a new technology, endowments, and transformation possibilities). The lack of knowledge of certain events causes the outcome of the adoption process to be indeterminate. In the context of network externalities, which firm or technology ultimately dominates cannot be deduced in advance, according to Arthur, because the actual outcome would likely be decided by a host of such small events.

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